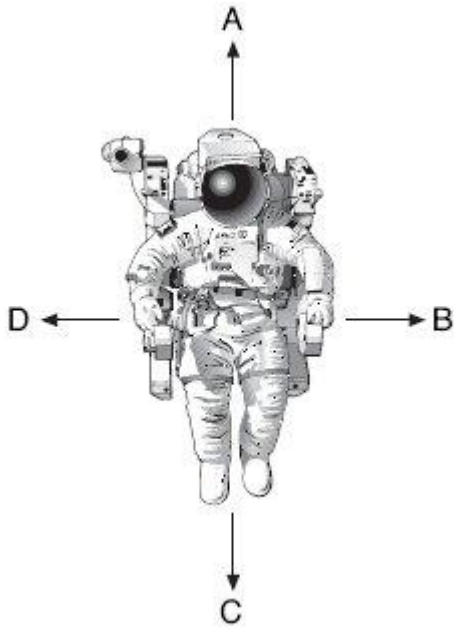
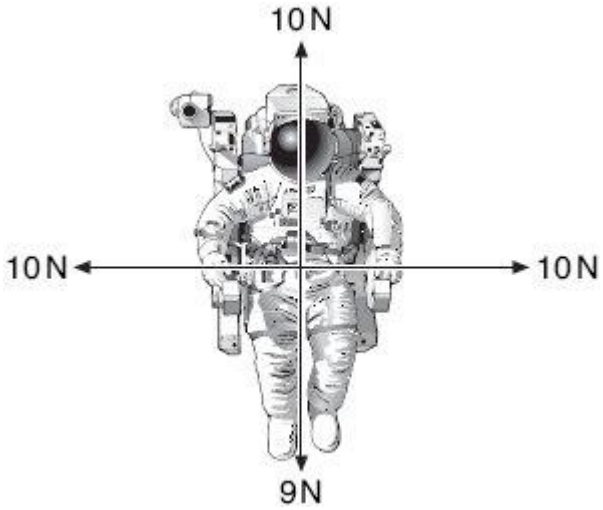
**Q1.** The drawing below shows an astronaut in space.
He has four small jets attached to his space suit.
These jets produce forces on the **astronaut** in the directions A, B, C and D.




(a) The drawing below shows the size and direction of four forces acting on the astronaut.



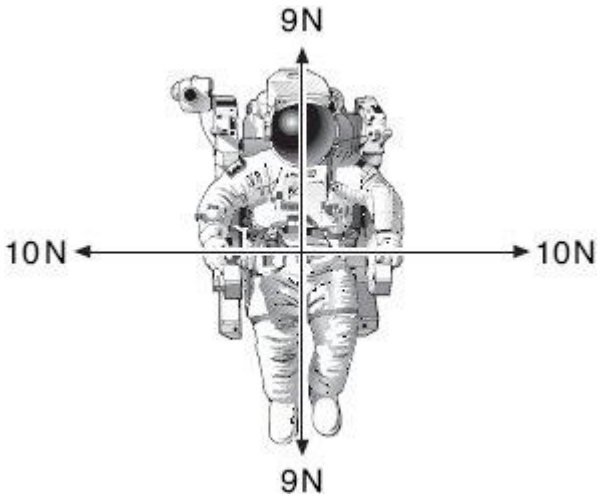

In which direction, A, B, C or D, will the astronaut move?

Give the letter.

.....

1 mark

(b) The drawing below shows the size and direction of four different forces acting on the astronaut.




What will happen to the astronaut when the jets produce these four forces?

.....

1 mark

Explain your answer.

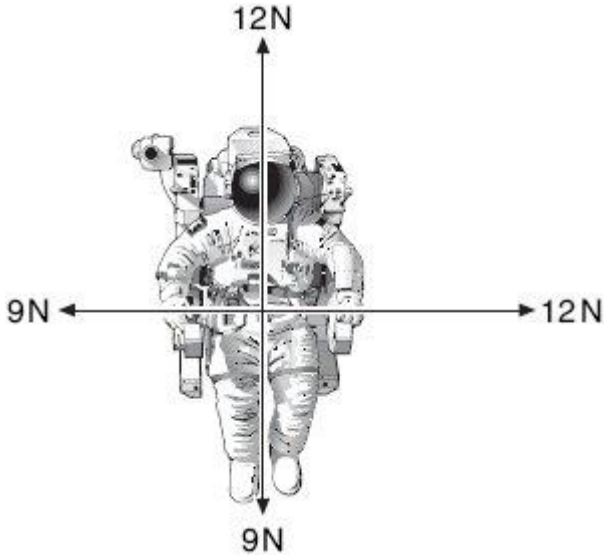
.....

.....

1 mark

(c) The drawing below shows the size and direction of four different forces acting on the astronaut.

Draw an arrow on the diagram below to show the direction in which he will move.




1 mark

maximum 4 marks

## Q1.

(a) A

**1 (L5)**

(b) any **one** from

- he will remain stationary
  *accept 'he floats'*
- he will continue moving at a constant speed
  *accept 'nothing'*

**1 (L6)**

any **one** from

- there is no net force
- the pairs of forces are equal
  *accept 'all the forces cancel out'*
  *accept 'they cancel each other out'*
  *accept 'the forces are balanced'*
  *'the forces are equal' is insufficient*

**1 (L6)**

(c) 

*accept any arrow drawn going up and to the right*

**1 (L6)**

**[4]**